=== UNIT 6: ANALYTICAL SPECTROSCOPY: UV-VISIBLE & INFRARED ===

>> 6.1 Fundamentals of Electromagnetic Radiation & Beer-Lambert Law

Analytical spectroscopy studies the interaction between electromagnetic radiation and molecular matter.
* Electromagnetic Spectrum: Radio -> Microwave (rotational) -> Infrared (vibrational) -> Visible & UV (electronic) -> X-ray (core electronic) -> Gamma-ray (nuclear).
- Planck-Einstein Relation: E = h * nu = h * c / lambda = h * c * nu_bar, where nu_bar = 1/lambda is wave number in cm^-1.
* Beer-Lambert's Law: When a beam of monochromatic radiation passes through an absorbing medium, the fractional decrease in radiant intensity with path length and concentration is constant:
- dI / I = k * c * db.
Integrating over path length l: ln(I_0 / I) = k' * c * l => log10(I_0 / I) = epsilon * c * l.
- Absorbance (A / Optical Density): A = log10(I_0 / I) = - log10(T) = epsilon * c * l,
where epsilon = Molar Absorptivity (Extinction Coefficient in L/mol*cm), c = concentration (mol/L), and l = path length (cm).
- Transmittance (T): T = I / I_0; %T = (I / I_0) * 100.
* Limitations: Deviations occur at high concentrations (>0.01 M) due to electrostatic molecular interactions, chemical equilibria (dimerization, pH shifts), and polychromatic stray light.

>> 6.2 Ultraviolet-Visible (UV-Vis) Spectroscopy

* Electronic Transitions: Absorption of UV (200-400 nm) or Visible (400-800 nm) photons excites valence electrons from bonding/non-bonding orbitals to anti-bonding orbitals:
1. sigma -> sigma* (Far UV, <200 nm, saturated hydrocarbons).
2. n -> sigma* (150-250 nm, saturated compounds with heteroatoms: O, N, S, halogens).
3. pi -> pi* (200-400 nm, unsaturated compounds, conjugated polyenes; intense absorption: epsilon > 10,000).
4. n -> pi* (250-400 nm, carbonyls, nitro, azo groups; symmetry-forbidden, low intensity: epsilon ~ 10-100).
* Chromophores and Auxochromes:
- Chromophore: Isolated covalently bonded unsaturated group responsible for electronic absorption (e.g., C=C, C=O, -N=N-, -NO2).
- Auxochrome: Saturated group with unshared electron pairs that does not absorb UV radiation itself, but when attached to a chromophore, deepens color and increases absorption intensity (e.g., -OH, -NH2, -OCH3, -SH).
* Spectral Shifts:

- Bathochromic (Red) Shift: Shift of absorption maximum (lambda_max) to longer wavelengths (lower energy), caused by conjugation or auxochrome attachment.
- Hypsochromic (Blue) Shift: Shift of lambda_max to shorter wavelengths (higher energy), caused by solvent polarity or protonation.
- Hyperchromic Effect: Increase in molar absorptivity (intensity). Hypochromic Effect: Decrease in intensity.

>> 6.3 Infrared (IR) Spectroscopy

* Principle: Absorption of IR radiation (4,000 - 400 cm^-1) excites molecular vibrations (stretching and bending). Mandatory Selection Rule: A molecular vibration is IR-active if and only if it produces a periodic change in the molecular dipole moment: d(mu)/dq != 0. Homonuclear diatomics (N2, O2, H2) are IR-inactive; heteronuclear molecules (HCl, CO, CO2, H2O) are IR-active.
* Hooke's Law for Vibrational Wave Number:
nu_bar = [1 / (2 * pi * c)] * sqrt(k / mu),
where k = force constant of the chemical bond (dyne/cm or N/m: k_single ~ 5*10^5, k_double ~ 10*10^5, k_triple ~ 15*10^5 dyne/cm), and mu = reduced mass = (m1 * m2) / (m1 + m2).
* Characteristic IR Functional Group Regions:
- O-H / N-H stretching: 3,200 - 3,600 cm^-1 (broad for hydrogen-bonded OH).
- C-H stretching: 2,850 - 3,100 cm^-1.
- C#C / C#N (Triple bond) stretching: 2,100 - 2,260 cm^-1.
- C=O (Carbonyl) stretching: 1,650 - 1,750 cm^-1 (very sharp and intense).
- C=C (Alkene/Aromatic) stretching: 1,600 - 1,680 cm^-1.
- Fingerprint Region: 1,400 - 400 cm^-1 (unique complex pattern for molecule identification).

[SOLVED PROBLEM 1: Beer-Lambert Law Molar Absorptivity and Concentration Analysis]

Problem: A 1.25 * 10^-4 M solution of an active biotechnology enzyme cofactor exhibits an absorbance of A = 0.625 at lambda_max = 280 nm in a 1.0 cm path-length quartz cuvette. Calculate: (1) Molar absorptivity epsilon in L/mol*cm, (2) Percentage transmittance (%T), (3) Concentration of an unknown solution of the same cofactor showing A = 0.250 in the same cell.
Solution Step 1: Calculate Molar Absorptivity epsilon:
A = epsilon * c * l => epsilon = A / (c * l)
epsilon = 0.625 / [(1.25 * 10^-4 mol/L) * (1.0 cm)] = 0.625 / 1.25 * 10^-4 = 5,000 L/mol*cm.
Solution Step 2: Calculate Percentage Transmittance (%T):
A = - log10(T) => T = 10^{-A} = 10^{-0.625} = 0.2371.
%T = T * 100 = 23.71%.
Solution Step 3: Determine Concentration of Unknown Solution:

c_unknown = A_unknown / (epsilon * l) = 0.250 / [5,000 * 1.0] = 0.000050 mol/L = 5.0 * 10^-5 M (50 microM).

[SOLVED PROBLEM 2: Hooke's Law Wave Number Calculation for Carbonyl Bond (C=O)]

Problem: Calculate the fundamental vibrational stretching frequency (wave number nu_bar in cm^-1) of a carbonyl bond (C=O) using Hooke's Law. Given: Force constant k = 12.0 * 10^5 dyne/cm; atomic masses: Carbon = 12.0 g/mol, Oxygen = 16.0 g/mol; velocity of light c = 3.0 * 10^10 cm/sec; Avogadro's number N_A = 6.022 * 10^23.

Solution Step 1: Calculate Atomic Masses in grams:

m_C = 12.0 / (6.022 * 10^23) = 1.993 * 10^-23 g.

m_O = 16.0 / (6.022 * 10^23) = 2.657 * 10^-23 g.

Solution Step 2: Compute Reduced Mass mu:

mu = (m_C * m_O) / (m_C + m_O) = [(1.993 * 2.657) * 10^-46] / [(1.993 + 2.657) * 10^-23]

mu = (5.295 * 10^-46) / (4.650 * 10^-23) = 1.1388 * 10^-23 grams.

Solution Step 3: Apply Hooke's Law Equation:

nu_bar = [1 / (2 * pi * c)] * sqrt(k / mu)

2 * pi * c = 2 * 3.14159 * (3.0 * 10^10 cm/s) = 1.88496 * 10^11 cm/s.

k / mu = (12.0 * 10^5 dyne/cm) / (1.1388 * 10^-23 g) = 1.0537 * 10^29 s^-2.

sqrt(k / mu) = sqrt(1.0537 * 10^29) = 3.246 * 10^14 s^-1.

nu_bar = (3.246 * 10^14) / (1.88496 * 10^11) = 1,722.0 cm^-1.

Conclusion: The calculated stretching wave number (1,722 cm^-1) matches the experimental carbonyl absorption range (1,680-1,750 cm^-1).

[SOLVED PROBLEM 3: Isotopic Substitution Effect on Vibrational Frequency]

Problem: The stretching frequency of a C-H bond is observed at 2,900 cm^-1. Calculate the expected stretching frequency of the corresponding C-D (deuterated) bond, assuming the bond force constant remains unchanged. (Atomic weights: C = 12, H = 1, D = 2).

Solution Step 1: Ratio of Reduced Masses:

mu_{CH} = (12 * 1) / (12 + 1) = 12 / 13 = 0.9231 amu.

mu_{CD} = (12 * 2) / (12 + 2) = 24 / 14 = 1.7143 amu.

Solution Step 2: Frequency Relationship:

nu_bar_{CD} / nu_bar_{CH} = sqrt(mu_{CH} / mu_{CD})

Ratio = sqrt(0.9231 / 1.7143) = sqrt(0.53846) = 0.7338.

Solution Step 3: Compute Deuterated Frequency:

nu_bar_{CD} = 2,900 cm^-1 * 0.7338 = 2,128.0 cm^-1.

Conclusion: Deuteration induces an 800 cm^-1 isotope shift into the infrared silent region.

[SOLVED PROBLEM 4: Multi-Component Spectrophotometric Analysis]

Problem: Two biochemical substances X and Y are present in a mixture. In a 1.0 cm cell at lambda1 (260 nm): epsilon_X = 15,000 and epsilon_Y = 3,000 L/mol*cm. At lambda2 (280 nm): epsilon_X = 4,000 and epsilon_Y = 12,000 L/mol*cm. The mixture exhibits absorbances of A_260 = 0.900 and A_280 = 0.800. Calculate concentrations C_X and C_Y.
Solution Step 1: Set Up Simultaneous Absorbance Equations:
At 260 nm: A_260 = (15,000 * C_X) + (3,000 * C_Y) = 0.900.
At 280 nm: A_280 = (4,000 * C_X) + (12,000 * C_Y) = 0.800.
Solution Step 2: Simplify Equations:
Divide Eq 1 by 3,000: 5 * C_X + C_Y = 0.000300 => C_Y = 0.000300 - 5 * C_X.
Divide Eq 2 by 4,000: C_X + 3 * C_Y = 0.000200.
Solution Step 3: Substitute C_Y into Eq 2:
C_X + 3 * (0.000300 - 5 * C_X) = 0.000200
C_X + 0.000900 - 15 * C_X = 0.000200 => -14 * C_X = -0.000700.
C_X = 0.000700 / 14 = 5.0 * 10^-5 M (50 microM).
Solution Step 4: Solve for C_Y:
C_Y = 0.000300 - 5 * (5.0 * 10^-5) = 0.000300 - 0.000250 = 5.0 * 10^-5 M (50 microM).
Final Answer: C_X = 5.0 * 10^-5 M, C_Y = 5.0 * 10^-5 M.

>> 6.4 University Review & Examination Questions

* Q1: State and derive Beer-Lambert’s Law, giving its mathematical formula, units, and practical analytical limitations.
Answer: Derivation: Consider a parallel beam of monochromatic light of intensity I passing through an absorbing solution of concentration c. Lambert’s law states that the rate of decrease of intensity with thickness db is proportional to initial intensity: -dI/db = k1*I. Beer’s law states that rate of decrease with concentration is proportional to intensity: -dI/dc = k2*I. Combining both: -dI/I = k*c*db. Integrating between limits I_0 (entering intensity at b=0) and I (transmitted intensity at path length l): int_{I_0}^I (-dI/I) = k*c*int_0^l db => ln(I_0 / I) = k*c*l. Converting to base-10 logarithm: log10(I_0 / I) = (k / 2.303)*c*l = epsilon*c*l. Absorbance A = epsilon*c*l. Units: l is in cm, c is in mol/L, and epsilon is molar absorptivity in L/(mol*cm). Limitations: (1) True linearity holds only at low concentrations (<0.01 M); at high concentrations, electrostatic intermolecular interactions alter refractive index. (2) Non-monochromatic stray light. (3) Chemical equilibria: changes in association, dissociation, or tautomerism alter absorbing species.
* Q2: Explain the selection rules and fundamental vibrational modes (stretching and bending) in Infrared (IR) spectroscopy.
Answer: Selection Rule: A molecular vibration absorbs infrared radiation if and only if it produces a periodic oscillation in the permanent molecular electric dipole moment: (d mu / dq) != 0. Homonuclear diatomic molecules (N2, O2) have zero dipole moment in all vibrational states and are completely IR-inactive. Heteronuclear molecules (HCl, CO) change dipole during stretching and are strongly IR-active. Degrees of Freedom: A

non-linear molecule containing N atoms has 3N - 6 fundamental vibrational modes; a linear molecule has 3N - 5 fundamental modes. Vibrational Modes: (1) Stretching Vibrations: Rhythmic movement along the chemical bond axis altering interatomic distance: Symmetrical stretching (bonds expand and contract in phase) and Asymmetrical stretching (one bond expands while the other contracts). (2) Bending (Deformation) Vibrations: Alter the bond angle between two bonds: Scissoring (movement toward each other in-plane), Rocking (structural swinging in-plane), Wagging (swinging up and down out-of-plane), and Twisting (rotation around bond out-of-plane).

>> 6.5 Key Takeaways and Revision Points

* Beer-Lambert’s Law: A = epsilon * c * l relates optical absorbance to concentration.
* UV-Vis spectroscopy measures electronic transitions (pi->pi*, n->pi*); auxochromes cause bathochromic (red) shifts.
* IR spectroscopy requires a net dipole moment change during molecular vibration.
* Hooke’s law: wave number nu_bar = (1/2*pi*c)*sqrt(k/mu); carbonyl (C=O) absorbs intensely at ~1,700 cm^-1.